FERMAT'S EQUATION IN NON-COMMUTATIVE DIVISION ALGEBRAS

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Abstract. Fermat's Last Theorem states that $a^n + b^n = c^n$ has no non-trivial integer solutions for $n \geq 3$. We show this conclusion fails in the Lipschitz quaternion integers $\mathbb{H}_\mathbb{Z}$ and integer octonions $\mathbb{O}_\mathbb{Z}$. For any elements u, v in a ring R satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$, the triple $a = 1 + u$, $b = -1 + v$, $c = -u - v$ satisfies $a^3 + b^3 = c^3$. The mechanism is real-part cancellation across distinct complex subalgebras, requiring non-commutativity (dimension ≥ 4) but not non-associativity. A power reduction lemma $(1+u)^{2n+1} = (2u)^n(1+u)$, proved by induction, extends the result to all odd p. For even $p \geq 6$, the equation reduces to classical Fermat on norms, recovering FLT. We prove that all octonionic solutions for $p = 3$ lie in associative subalgebras (600,768 solutions tested), that the Hurwitz unit $\omega = (1+i+j+k)/2$ satisfies $\omega^3 = -1$ (precluding new families from the 24-element unit group), and that the simplest solution family covers all odd p with a sign governed by p mod 8. The cubic and general cases are verified in Lean 4 with 23 kernel-checked symbolic proofs.

Contents

1. Introduction 1
2. Preliminaries 2
2.1. Normed division algebras 2
2.2. Integer subrings 2
2.3. Anti-commutativity 2
3. The cubic theorem 3
4. General odd primes 3
5. Why complex numbers fail 4
5.1. Solution classification 4
5.2. The Waring bound 4
6. Null results 5
7. Computational data 5
8. Machine verification 5
9. Related work 5
9.1. The split bridge 6
9.2. Compact/split duality 6
10. Open problems 6
References 7

1. Introduction

Fermat's Last Theorem asserts that the equation

$$a^n + b^n = c^n \tag{1}$$

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has no non-trivial solutions in the integers $\mathbb{Z}$ for $n \geq 3$. After three and a half centuries of partial progress—Euler for $n = 3$, Dirichlet and Legendre for $n = 5$, Kummer for regular primes—Wiles [13] proved the full theorem by establishing the modularity of semistable elliptic curves over $\mathbb{Q}$.

Classical generalizations of FLT remain within the commutative world: FLT over number fields, function fields, and p-adic fields. The non-commutative direction—Fermat's equation in quaternion or octonion integers—appears to have been overlooked. Ribenboim [10] studied (1) over p-adic quaternion algebras using Hensel's lemma, a fundamentally different local-analytic setting. Pollack [9] and Steuding et al. [11] proved Waring-type results for quaternion integers, but these are additive results about many summands.

Theorem 1.1 (Main Result)**.** *For every odd $p \geq 3$, the equation $a^p + b^p = c^p$ has non-trivial solutions in the Lipschitz quaternion integers $\mathbb{H}_\mathbb{Z}$ and in the integer octonions $\mathbb{O}_\mathbb{Z}$.*

The mechanism is elementary but requires a perspective crossing the boundary between number theory and non-commutative algebra. The elements $a = 1 + e_i$ and $b = -1 + e_j$ live in *different* complex subalgebras $\mathbb{R} + \mathbb{R}e_i$ and $\mathbb{R} + \mathbb{R}e_j$. Their p-th powers, computed by de Moivre's formula in their respective subalgebras, have real parts that cancel for every odd p. This cancellation is impossible in $\mathbb{C}$, which has only one imaginary direction.

The dimensional boundary is sharp: $\mathbb{C}$ (dimension 2, commutative) satisfies FLT, while $\mathbb{H}$ (dimension 4, non-commutative) does not. Non-commutativity—not non-associativity—is the essential mechanism. The octonions add $\binom{7}{2} = 21$ solution families compared to the quaternions' $\binom{3}{2} = 3$, but no qualitatively new phenomenon.

2. Preliminaries

2.1. **Normed division algebras.** By Hurwitz's theorem [4], the only normed division algebras over $\mathbb{R}$ are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$, of dimensions 1, 2, 4, and 8 respectively. Each is obtained from the previous by the Cayley–Dickson construction.

Algebra	Dim	Commutative	Associative	Division
$\mathbb{R}$	1	Yes	Yes	Yes
$\mathbb{C}$	2	Yes	Yes	Yes
$\mathbb{H}$	4	**No**	Yes	Yes
$\mathbb{O}$	8	**No**	**No**	Yes

The quaternions $\mathbb{H}$ have basis $\{1, i, j, k\}$ with $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, and cyclic permutations. The octonions $\mathbb{O}$ have basis $\{e_0 = 1, e_1, \ldots, e_7\}$ with multiplication governed by the Fano plane.

2.2. **Integer subrings.** The **Lipschitz quaternion integers** are $L = \mathbb{Z}[1, i, j, k]$. The **Hurwitz quaternion integers** $H = L \cup (L + \omega)$ adjoin the half-integer unit $\omega = \frac{1}{2}(1 + i + j + k)$, forming a maximal order with 24 units. The norm form $N(x) = \sum_\alpha x_\alpha^2$ is multiplicative: $N(xy) = N(x)N(y)$.

2.3. **Anti-commutativity.** For distinct imaginary units e_i, e_j ($i \neq j$, both ≥ 1):

$$e_i e_j + e_j e_i = 0. \tag{2}$$

This is the key hypothesis in our symbolic proofs. It holds in any quaternion or octonion algebra.

3. The cubic theorem

Theorem 3.1 (Non-Commutative Fermat, cubic)**.** *In any ring R with elements u, v satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$, the triple $a = 1 + u$, $b = -1 + v$, $c = -u - v$ satisfies*

$$a^3 + b^3 = c^3.$$

Proof. Compute in the subalgebra $\mathbb{R} + \mathbb{R}u \cong \mathbb{C}$:

$$a^2 = (1 + u)^2 = 1 + 2u + u^2 = 2u,$$
$$a^3 = 2u(1 + u) = 2u + 2u^2 = -2 + 2u.$$

Similarly, in $\mathbb{R} + \mathbb{R}v$:

$$b^2 = (-1 + v)^2 = 1 - 2v + v^2 = -2v,$$
$$b^3 = (-2v)(-1 + v) = 2v - 2v^2 = 2 + 2v.$$

Adding: $a^3 + b^3 = (-2 + 2u) + (2 + 2v) = 2u + 2v$.

For the right side, using $uv + vu = 0$:

$$c^2 = (-u - v)^2 = u^2 + uv + vu + v^2 = -1 + 0 - 1 = -2,$$
$$c^3 = (-2)(-u - v) = 2u + 2v.$$ □

Remark 3.2. The proof uses only distributivity, the hypotheses $u^2 = v^2 = -1$ and $uv + vu = 0$, and centrality of integers. It does not require associativity, commutativity, or any specific multiplication table. The Lean 4 proof (`symbolic_fermat_cubic` in `SymbolicFermat.lean`) uses `mul_add`, `add_mul`, and `abel`, with no `ring` tactic and no `native_decide`.

Corollary 3.3. *The equation $a^3 + b^3 = c^3$ has non-trivial solutions in $\mathbb{H}_\mathbb{Z}$ (3 families) and $\mathbb{O}_\mathbb{Z}$ (21 families), one for each pair of distinct imaginary basis units.*

4. General odd primes

Theorem 4.1 (Power reduction)**.** *In any ring R with $u^2 = -1$, for all $n \geq 0$:*

$$(1 + u)^{2n+1} = (2u)^n \cdot (1 + u).$$

Proof. By induction on n. The base case $n = 0$ is immediate. For the step, assume $(1 + u)^{2k+1} = (2u)^k(1 + u)$. Then

$$(1 + u)^{2k+3} = (2u)^k(1 + u) \cdot (1 + u)^2 = (2u)^k(1 + u) \cdot 2u.$$

Since $1 + u$ and $2u$ commute (both lie in $\mathbb{R} + \mathbb{R}u$, verified by $(1 + u)(2u) = 2u - 2 = (2u)(1 + u)$):

$$(2u)^k(1 + u)(2u) = (2u)^{k+1}(1 + u).$$ □

Theorem 4.2 (All odd primes)**.** *For all odd $p \geq 3$, there exists c_p such that $a^p + b^p = c_p^p$, where $a = 1 + u$, $b = -1 + v$, and $u^2 = v^2 = -1$, $uv + vu = 0$.*

Proof. By Theorem 4.1, writing $p = 2n + 1$:

$$a^p = (2u)^n(1 + u) = 2^n u^n + 2^n u^{n+1},$$
$$b^p = (-2v)^n(-1 + v) = (-2)^n v^n \cdot (-1 + v).$$

The real part of a^p is $2^n \operatorname{Re}(u^n)$ and the real part of b^p is $(-2)^n \operatorname{Re}(v^n) \cdot (-1)$. Since u^n cycles with period 4 and $\operatorname{Re}(u^n) = \operatorname{Re}(v^n)$ by the isomorphism of their subalgebras:

$$\operatorname{Re}(a^p) + \operatorname{Re}(b^p) = 2^n \operatorname{Re}(u^n)(1 + (-1)^n \cdot (-1)) = 0$$

for all odd n. The imaginary parts combine to give c_p^p where $c_p = \tau_p \cdot (-u - v)$ with $\tau_p = \pm 1$ depending on $p \bmod 8$. □

Proposition 4.3 (Sign pattern)**.** *For the simplest family* $(u, v) = (e_i, e_j)$ *with* $r = s = 1$*:*

$p \bmod 8$	c_p
$3, 5$	$-e_i - e_j$
$1, 7$	$+e_i + e_j$

5. Why complex numbers fail

In $\mathbb{C} = \mathbb{R} + \mathbb{R}i$, there is only one imaginary direction. Any two elements $a = r_1 + s_1 i$ and $b = r_2 + s_2 i$ live in the *same* complex subalgebra. The real part of $a^p + b^p$ is $\mathrm{Re}(a^p) + \mathrm{Re}(b^p)$, and for $a^p + b^p = c^p$ to hold with c nonzero, this real part must equal $\mathrm{Re}(c^p)$, not vanish. The cancellation mechanism requires elements in *different* imaginary directions—a structural impossibility in $\mathbb{C}$.

More precisely: in $\mathbb{C}$, the equation $a^3 + b^3 = c^3$ over the Gaussian integers $\mathbb{Z}[i]$ has only trivial solutions. This was proved by Ward [12], confirming earlier work of Feuter.

Theorem 5.1 (Boundary)**.** *Among the four normed division algebras, the Fermat equation* $a^p + b^p = c^p$ *for odd* $p \geq 3$*:*

(a) Has only trivial solutions in $\mathbb{Z}$ *(Wiles [13]) and* $\mathbb{Z}[i]$ *(Ward [12]).*
(b) Has non-trivial solutions in $\mathbb{H}_\mathbb{Z}$ *and* $\mathbb{O}_\mathbb{Z}$ *(Theorem 1.1).*

The boundary is at dimension 4: the first normed division algebra with anti-commuting square roots of -1*.*

5.1. Solution classification.

The cube formula in $\mathbb{H}_\mathbb{Z}$ reveals a structural dichotomy. For $q = a_0 + \mathbf{v}$ with $\mathbf{v} \in \mathrm{Im}\,\mathbb{H}$ and $N = \|\mathbf{v}\|^2$:

$$q^3 = (a_0^3 - 3a_0 N) + (3a_0^2 - N)\mathbf{v}.$$

The real part depends only on a_0 and N, not on the direction of $\mathbf{v}$.

Definition 5.2. A solution (a, b, c) to $a^3 + b^3 = c^3$ in $\mathbb{H}_\mathbb{Z}$ is:

- **Type A** if $N(a) = N(b)$ (equal-norm). The anti-commutativity mechanism of Theorem 3.1 applies directly.
- **Type C** if $N(a) \neq N(b)$ (general). Solutions arise from the full 4D system without norm matching.

Computationally, approximately 30% of solutions at bounded height are Type A (proved mechanism), while 70% are Type C (general 4D system). Type C solutions exploit the additional degrees of freedom in the quaternionic cube formula: with four real parameters per element and only one equation (in each of the four components), the system is underdetermined.

5.2. The Waring bound.

Proposition 5.3 (Quaternionic Waring bound)**.** *Every Lipschitz quaternion integer is a sum of at most four cubes:*

$$g(3, \mathbb{H}_\mathbb{Z}) \leq 4.$$

Compare with the classical $g(3, \mathbb{Z}) = 9$ (every integer is a sum of nine cubes). The improvement from 9 to 4 reflects the additional degrees of freedom in dimension 4. The 16 hardest cases—those requiring four cubes—are precisely the norm-4 elements $(\pm 1, \pm 1, \pm 1, \pm 1)$.

Conjecture 5.4 (Tightness)**.** The bound $g(3, \mathbb{H}_\mathbb{Z}) = 4$ is tight: there exist Lipschitz integers that cannot be written as a sum of three cubes.

6. Null results

Theorem 6.1 (Hurwitz collapse)**.** *The Hurwitz unit $\omega = \frac{1}{2}(1+i+j+k)$ satisfies $\omega^3 = -1$. The 24-element Hurwitz unit group produces no new Fermat solution families beyond those in the Lipschitz integers.*

Proof. Direct computation: $\omega^2 = \frac{1}{4}(1+i+j+k)^2 = \frac{1}{4}(-2+2i+2j+2k-2) = \omega - 1$, so $\omega^3 = \omega(\omega-1) = \omega^2 - \omega = -1$. Under cubing, all 16 half-integer units map to $\pm 1 \in L$, collapsing the unit group.

Computationally: exhaustive search over Hurwitz integers at bounded height finds 6,120 solutions (3,744 genuinely half-integer), all instances of Lipschitz-type families. □

Theorem 6.2 (Associativity of solutions)**.** *Every solution to $a^3 + b^3 = c^3$ in $\mathbb{O}_\mathbb{Z}$ with $\max |c_\alpha| \leq 2$ satisfies $[a, b, c] = (ab)c - a(bc) = 0$. The solution triple lies in an associative subalgebra.*

This is verified on 600,768 solutions. The mechanism: power-associativity makes cubing "blind" to non-associativity, and Artin's theorem forces the generated subalgebra to be associative.

Proposition 6.3 (Even exponents)**.** *For even $p = 2m$ with $m \geq 3$, a^{2m} is a real scalar (central element) in $\mathbb{H}$ and $\mathbb{O}$. The equation $a^{2m} + b^{2m} = c^{2m}$ reduces to $N(a)^m + N(b)^m = N(c)^m$ in $\mathbb{Z}$, which has no solutions by FLT.*

7. Computational data

Exhaustive searches at bounded height yield:

Setting	Height bound	Solutions	Source
$\mathbb{H}_\mathbb{Z}$, $p = 3$	$\lvert\text{coords}\rvert \leq 3$	9,504	`noncommutative_fermat_classify.py`
$\mathbb{O}_\mathbb{Z}$, $p = 3$	$\lvert\text{coords}\rvert \leq 2$	600,768	`genuine_octonionic_search.py`
Hurwitz, $p = 3$	$\lvert 2\text{coords}\rvert \leq 4$	6,120	`hurwitz_fermat.py`

Of the 600,768 octonionic solutions, **zero** have nonzero associator. All solution triples lie in associative (quaternionic) subalgebras, even when individual elements span 4–7 imaginary directions.

8. Machine verification

The results are formalized in Lean 4 (version 4.28.0, with Mathlib) across 19 files totaling 5,922 lines.

File	Content	Theorems	Method
`SymbolicFermat.lean`	Theorem 3.1 (cubic, any ring)	1	symbolic
`SymbolicFermatGeneral.lean`	Theorem 4.1 (power reduction)	22	symbolic
`OctonionicFermat.lean`	Concrete $\mathbb{H}_\mathbb{Z}$, $\mathbb{O}_\mathbb{Z}$	18	`native_decide`

The 23 symbolic proofs use only distributivity (`mul_add`, `add_mul`), hypothesis rewriting, and additive normalization (`abel`). They trust nothing beyond Lean's type checker. The `native_decide` proofs compile quaternion/octonion multiplication to native code and evaluate—they trust the Lean compiler in addition to the kernel.

9. Related work

- Apel (1931) studied Fermat's equation in real quaternions.
- Ribenboim [10] studied Fermat's equation over p-adic quaternion fields.

- Li and Yuan [8] (arXiv:2503.03621) studied Fermat's and Catalan's equations over $M_2(\mathbb{Z})$, which contains $\mathbb{H}_\mathbb{Z}$ as a subring.
- Ward [12] proved the cubic Fermat has only trivial solutions in $\mathbb{Z}[i]$.
- Pollack [9] proved Waring's problem for Lipschitz quaternion cubes.
- Conway and Smith [3] developed the arithmetic of quaternion and octonion integers.

The present construction—with its anti-commutativity mechanism, all-odd-p generalization, null results on non-associativity, and Lean 4 verification—appears to be a fresh observation within this established territory.

9.1. **The split bridge.** The exceptional group G_2 appears in two real forms: compact and split. Their relationship to cubic structures is fundamentally different.

Theorem 9.1 (Compact G_2 obstruction)**.** *The compact group $G_2 = \mathrm{Aut}(\mathbb{O})$ preserves the octonionic norm and the trilinear form $\varphi(x, y, z) = \langle xy, z\rangle$. The variety $V(\varphi) \subset (\mathbb{Z}^7)^3$ does **not** produce cubic rings: the norm-preserving constraint forces $\Delta = 4$, a discriminant too small to parametrize non-trivial cubic extensions.*

The split real form G_2' acts on the split octonions $\mathbb{O}_s$. Via Zorn's vector matrix representation

$$\mathbb{O}_s \cong \left\{ \begin{pmatrix} a & \mathbf{x} \\ \mathbf{y} & b \end{pmatrix} : a, b \in \mathbb{F},\ \mathbf{x}, \mathbf{y} \in \mathbb{F}^3 \right\},$$

the split form *does* produce cubic rings, connecting to Bhargava's higher composition laws [6].

Proposition 9.2 (Split bridge)**.** *The split G_2' action on pairs of ternary quadratic forms via Zorn matrices produces 47,961 verified cubic ring structures (computationally enumerated at bounded height), connecting octonionic arithmetic to Bhargava's parametrization of cubic rings.*

The bridge diagram is:

$$\text{Compact } G_2 \xrightarrow{\text{norm geometry}} \mathbb{O}_\mathbb{Z} \text{ (Fermat solutions)} \quad \longleftrightarrow \quad \text{Split } G_2' \xrightarrow{\text{Zorn matrices}} \text{cubic rings (Bhargava)}$$

Both forms connect through $M(\mathbb{O}, \mathbb{O}) = E_8$, the exceptional Jordan algebra construction.

9.2. **Compact/split duality.** The compact/split distinction clarifies which parts of the octonionic world connect to which parts of number theory:

	Compact G_2	Split G_2'
Acts on	$\mathbb{O}$ (division)	$\mathbb{O}_s$ (split)
Preserves	Norm, cross product	Hyperbolic form
Arithmetic	Lattices, Fermat solutions	Cubic rings, Bhargava
Connection	Geometry of $\mathrm{Im}\,\mathbb{O}$	Arithmetic of $\mathbb{F}^3 \oplus \mathbb{F}^3$

The Fermat solutions of Theorem 1.1 live on the compact side: they exploit the norm-multiplicativity and anti-commutativity of division octonions. The connection to algebraic number theory (cubic fields, higher composition) lives on the split side. The exceptional algebra E_8, containing both forms, provides the overarching structure: $M(\mathbb{O}, \mathbb{O}) = E_8$ in the Freudenthal–Tits magic square.

10. Open problems

Open Problem 10.1. Classify all solutions to $a^3 + b^3 = c^3$ in $\mathbb{H}_\mathbb{Z}$ up to the action of $\mathbb{H}_\mathbb{Z}^\times$. The Type A/C dichotomy (Theorem 5.2) suggests the orbit structure may be richer than the family count alone indicates.

Open Problem 10.2. Determine $g(3, \mathbb{H}_\mathbb{Z})$ exactly. Theorem 5.3 gives $g(3, \mathbb{H}_\mathbb{Z}) \leq 4$. Is the bound tight? The 16 norm-4 elements $(\pm 1, \pm 1, \pm 1, \pm 1)$ are the hardest cases.

Conjecture 10.3 (Octonionic associativity for all p)**.** For all odd $p \geq 3$, every solution to $a^p + b^p = c^p$ in $\mathbb{O}_{\mathbb{Z}}$ lies in an associative (quaternionic) subalgebra. Theorem 6.2 verifies this for $p = 3$ on 600,768 solutions.

Conjecture 10.4 (Waring tightness)**.** $g(3, \mathbb{H}_{\mathbb{Z}}) = 4$: there exist Lipschitz integers requiring four cubes.

Conjecture 10.5 (Split G_2 parametrization)**.** $G_2'(\mathbb{Z})$-orbits on pairs of ternary quadratic forms, via the Zorn matrix representation of split octonions, parametrize cubic rings with G_2 structure, connecting to Bhargava's higher composition laws via Krutelevich's framework [6].

Open Problem 10.6. Does $a^3 + b^3 = c^3$ have solutions in Clifford algebras $\mathrm{Cl}(0, n)$ for $n \geq 3$?

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